清華大學 Tsinghua University

实验室科研探究课程

激光微纳制造
–调控电子的艺术

机械工程系
闫剑锋

李兆基科技大楼A107

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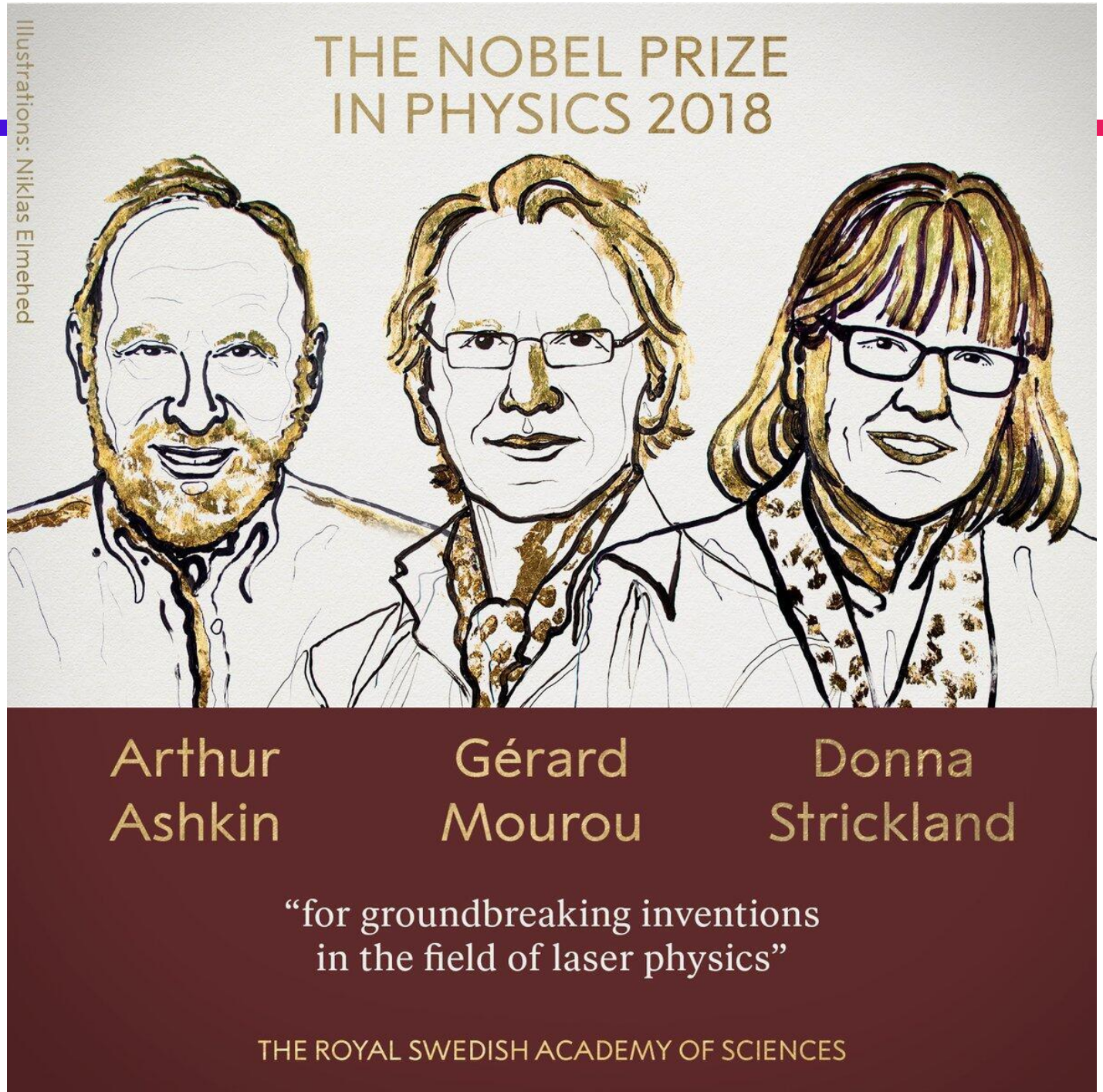
激光 （受激辐射光放大）

LASER Light Amplification by Stimulated Emission of Radiation

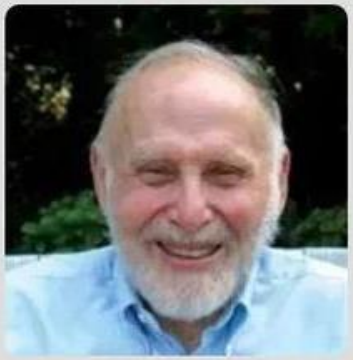
激光是20世纪以来继核能、电脑、半导体之后，人类的又一重大发明，被称为“最快的刀”、“最准的尺”、“最亮的光”。原子受激辐射的光，故名“激光”。

激光之所以被誉为神奇的光，是因为它有普通光所完全不具备的四大特性:

- 方向性好（激光准直、导向和测距）
- 亮度高 ——激光是当代最亮的光源，只有氢弹爆炸瞬间强烈的闪光才能与它相比拟。（激光打孔、切割、焊接和激光外科就是利用了这一特性）
- 单色性好 ——波长变化范围不到万分之一纳米
- 相干性好——光的频率单一


Illustrations: Niklas Elmehed
THE NOBEL PRIZE
IN PHYSICS 2018
Arthur Ashkin
Gérard Mourou
Donna Strickland
"for groundbreaking inventions
in the field of laser physics"
THE ROYAL SWEDISH ACADEMY OF SCIENCES

阿瑟·阿什金



阿瑟·阿什金（Arthur Ashkin）是美国物理学家，出生于1922年，曾工作于贝尔实验室。上个世纪60年代，他开展了运用激光操纵微粒的研究，最终导致了光镊的发明。

杰哈•莫罗



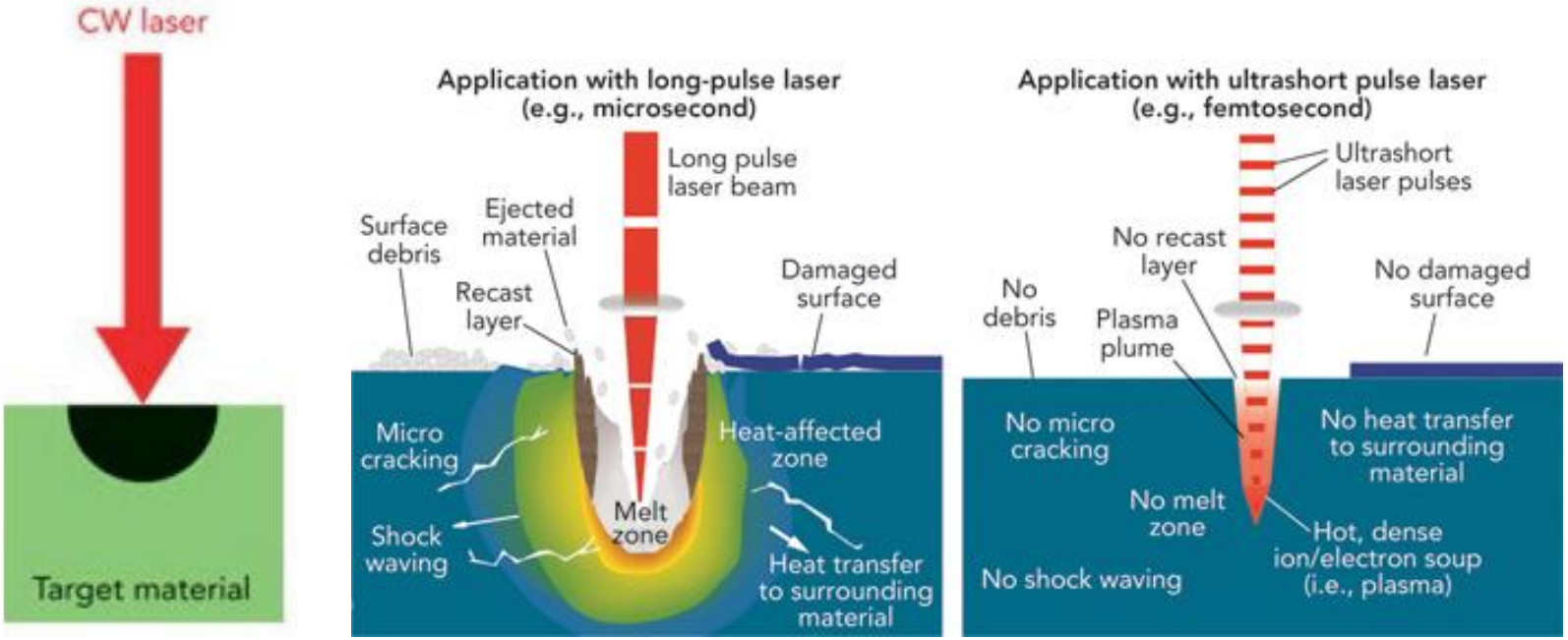
杰哈•莫罗（Gérard Mourou）是法国激光物理学家，现为密歇根大学教授。其开创了超快激光器的研究和应用领域。他是啁啾脉冲放大（CPA）技术的发明者，该技术能够将超短激光脉冲放大到非常高的功率。

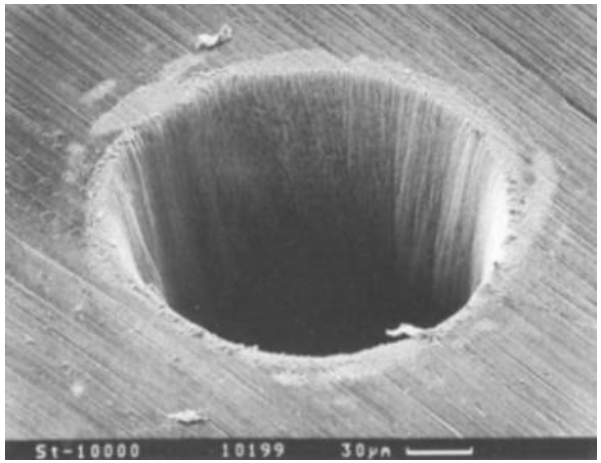
多娜·斯崔克兰

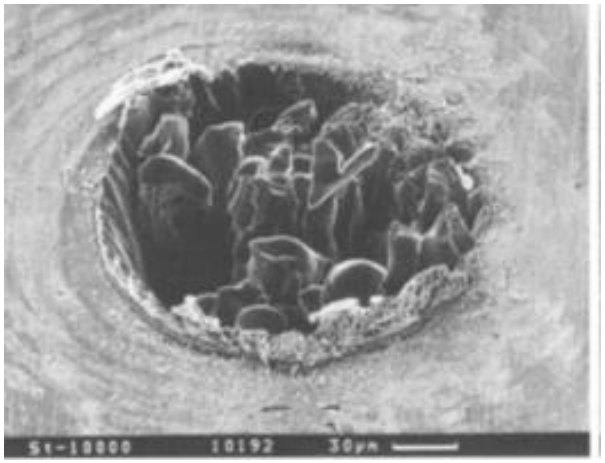


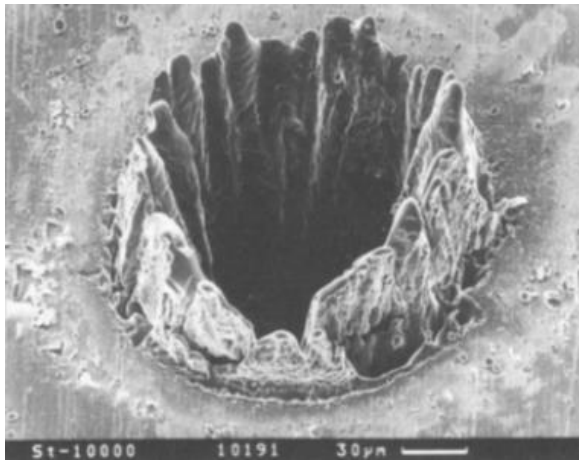
多娜·斯崔克兰（Donna Strickland）是加拿大物理学家，出生于1959年，现为滑铁卢大学副教授。其跟随杰哈•莫罗攻读博士学位期间与莫罗一起发明了啁啾脉冲放大技术。目前其研究方向仍为超快激光相关领域。

- Femtosecond laser (10^{-15} of a Second)
- Picosecond laser (10^{-12} of a Second)
- Nanosecond laser (10^{-9} of a Second)
- Microsecond laser (10^{-6} of a Second)
- Continuous wave laser

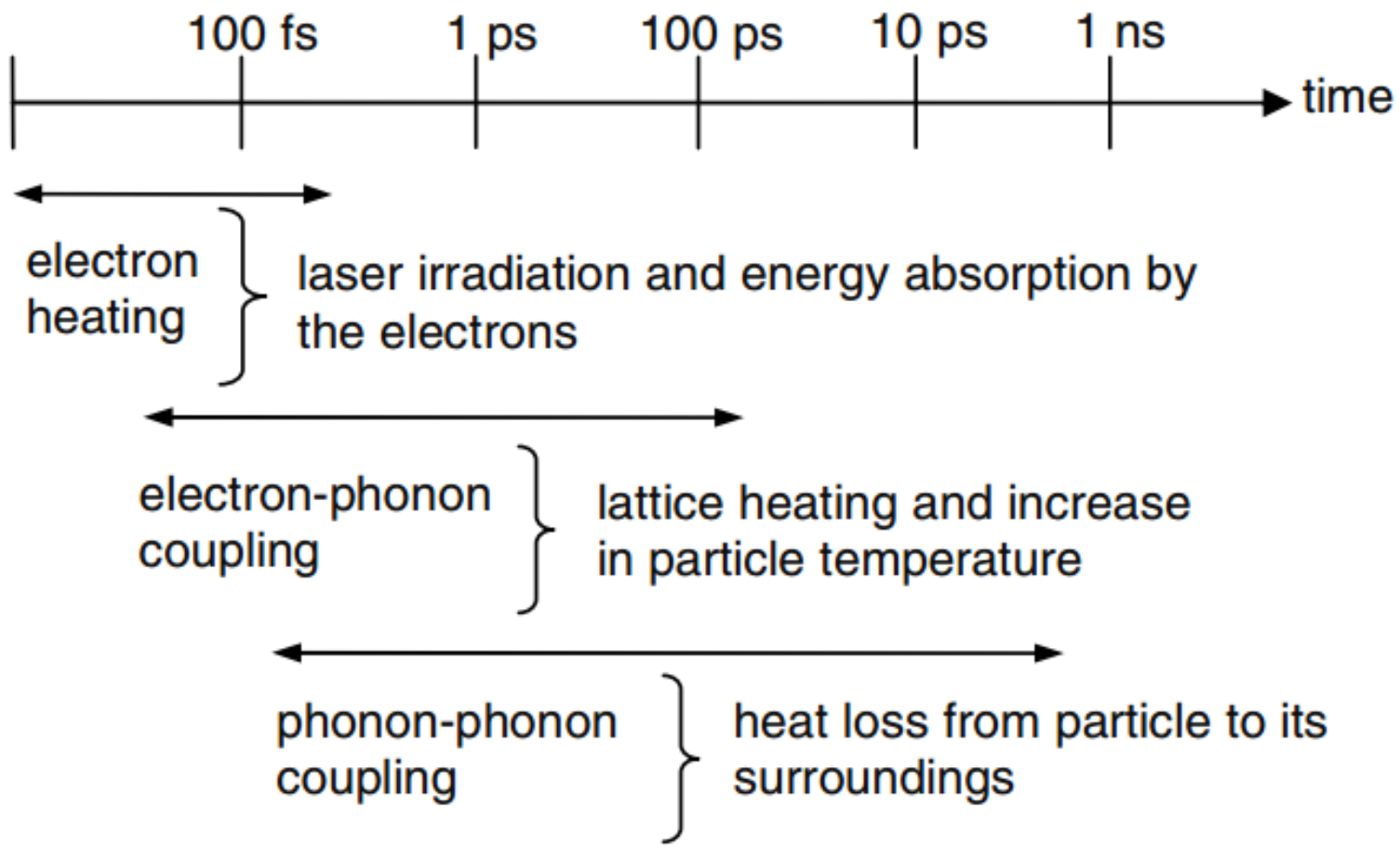






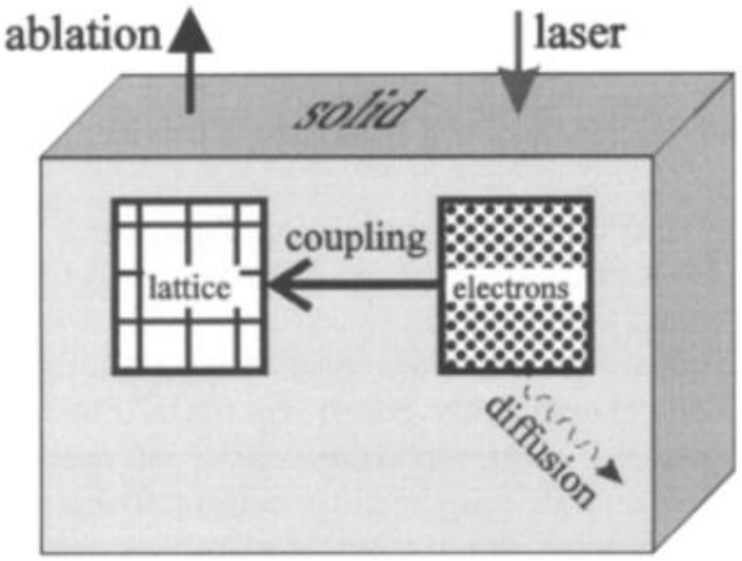


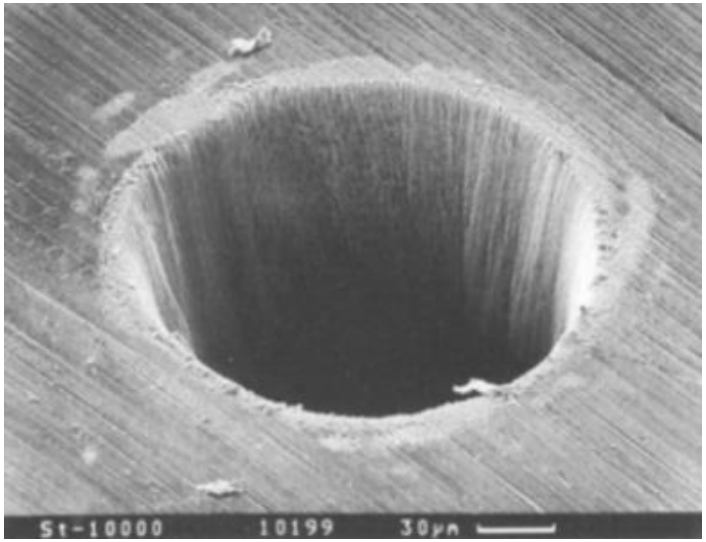
- **Femtosecond laser (10^{-15} of a Second)**
- **Picosecond laser (10^{-12} of a Second)**
- **Nanosecond laser (10^{-9} of a Second)**

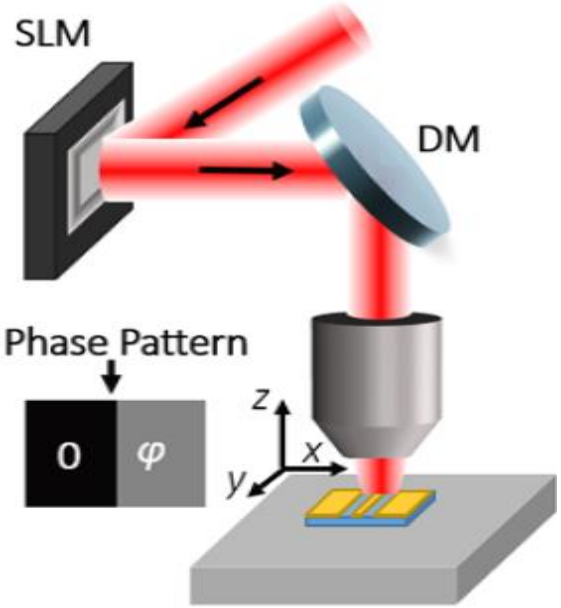


Time scales of the fundamental processes in ultrafast laser heating

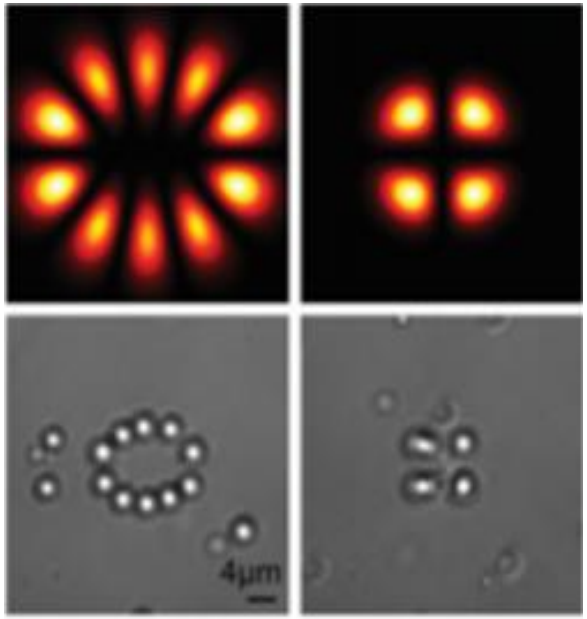
- **The electron-lattice energy relaxation time:$10^{-10} - 10^{-12} s$**
- **Laser energy absorption is completed before the lattice changes**



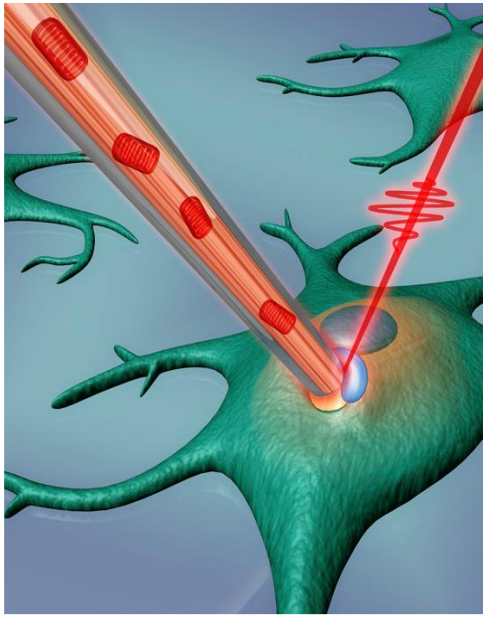




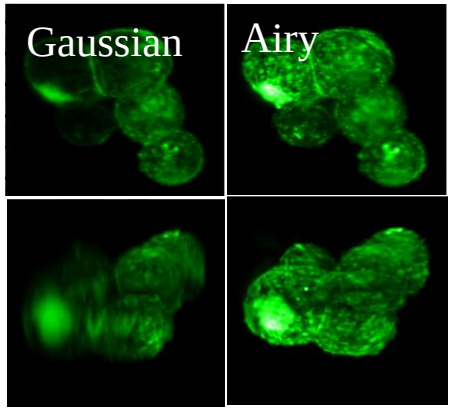
微纳制造



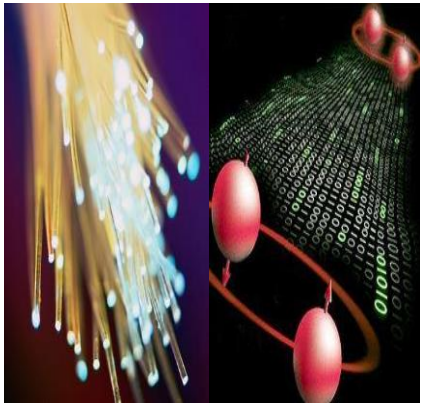
灵巧光镊



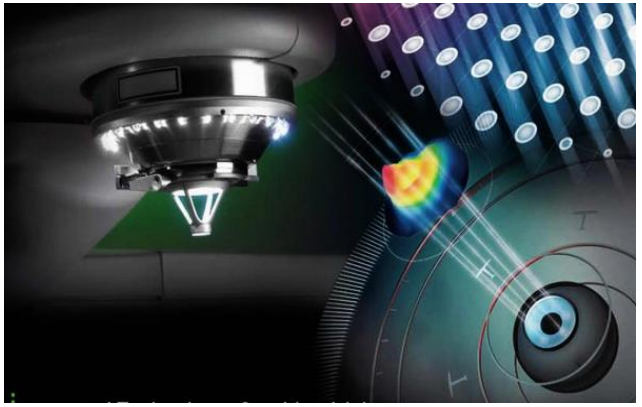
细胞精密手术



超分辨成像



量子通讯



飞秒激光手术

❑ 超快激光微纳制造

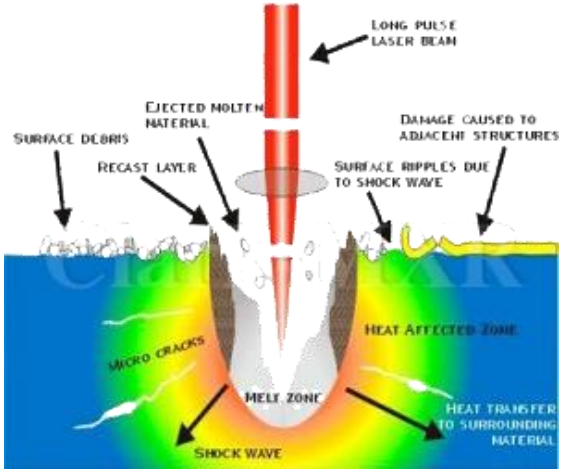
➢ 可实现极端条件制造

超快： 10^{-18}-10^{-11} s < 晶格升温时间

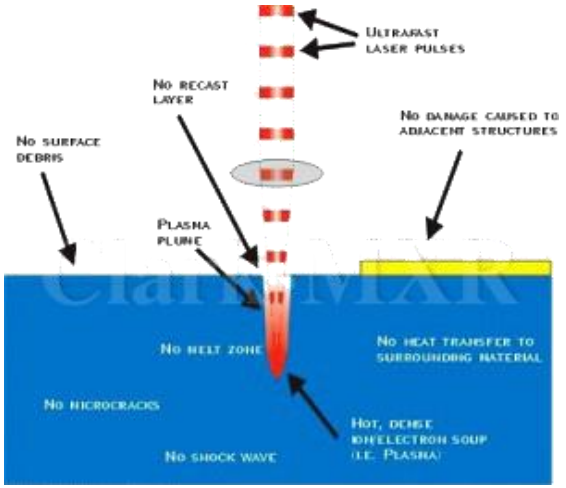
“冷加工”

超强： 10^{14}-10^{22}W/cm^2，**非线性**电离

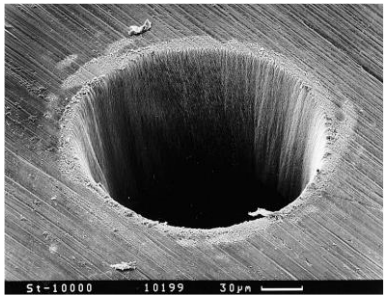
效果： 极小化重铸层、微裂纹、热影响区等种种缺陷态，大大提高加工精度



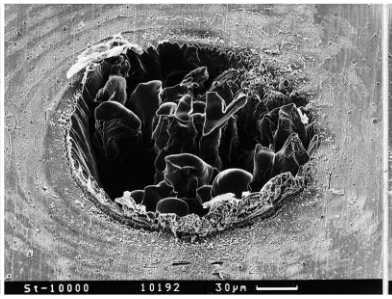
长脉冲激光与材料作用



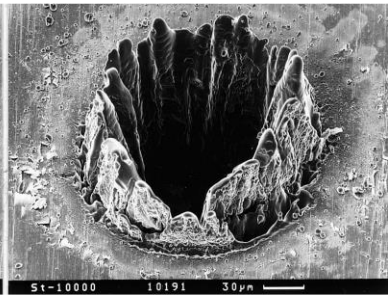
超短脉冲激光与材料作用



飞秒激光加工

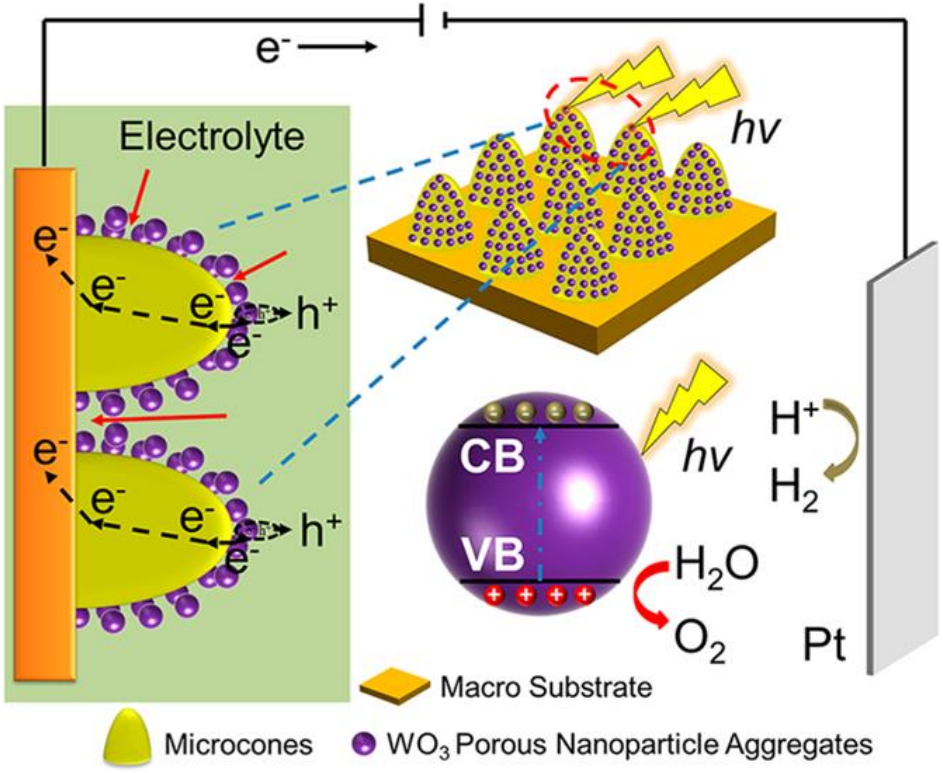


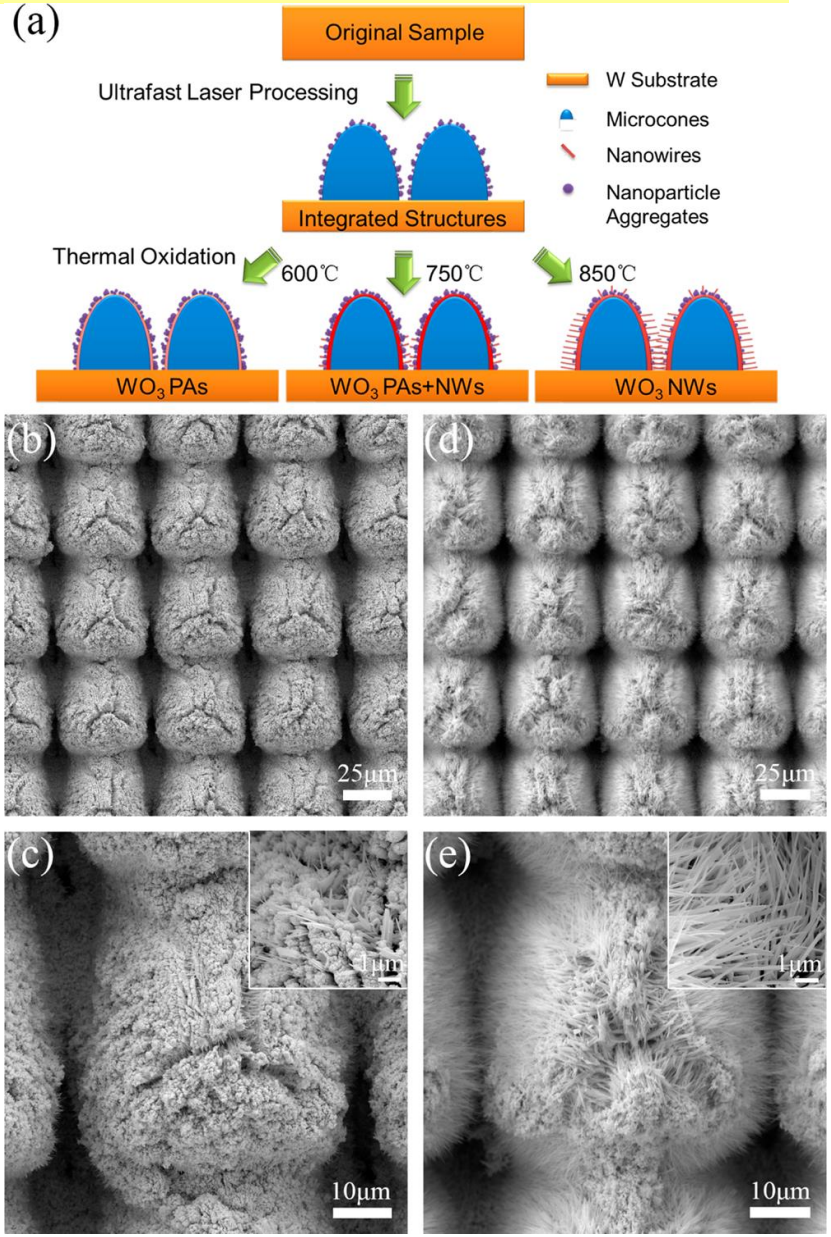
皮秒激光加工



纳秒激光加工

WO_3 Micro-Nano Architectures as Direct Photoanodes for Efficient Photoelectrochemical Water Splitting





Schematic of the hierarchical structure with porous WO_3 nanoparticle aggregates and the corresponding charge transport/transfer and electrolyte permeation processes.

谢谢大家!